FrameContact: Contact-Derived Task Routing for Antibody–Antigen Interfaces

Anonymous Authors

Abstract

Antibody–antigen interface prediction couples three related outputs: residue-pair contacts and their antibody-side paratope and antigen-side epitope marginals. A single jointly trained checkpoint, however, can be suboptimal for one output. We introduce FrameContact, a partner-conditioned 3D architecture that combines protein-language-model residue embeddings, local backbone frames, an explicit low-rank contact map, and interleaved bidirectional cross-attention. Under a fixed SAbDab2 development protocol, epoch-level diagnostics and a 2×2 task ablation show that the fully joint objective gives the strongest epitope result. The curves are consistent with late overfitting and do not support simple multitask competition. We therefore retain joint learning and change only the epitope readout: an epitope expert aggregates the two strongest predicted contact edges per antigen residue, and deterministic task routing uses this expert only for epitope while preserving contact and paratope from the joint baseline. Before full-development retraining, we freeze this route and evaluate its screen-stage checkpoints once on an antigen-cluster-disjoint development holdout of 156 complexes. All three seeds improve epitope MPC-AP (+0.0057, +0.0141, and +0.0089); a paired bootstrap over 49 antigen clusters gives a cluster-weighted, seed-averaged improvement of +0.0295 with 95% CI $[0.0126, 0.0474]$, conditional on the three trained seeds. We then retrain on the complete 2,565/549 development split, whose 549-complex validation set selects checkpoints; its routed three-seed validation median improves from 0.2324 to 0.2375 (+0.0051), with contact and paratope unchanged by construction. The fresh156 result validates the frozen routing decision, not the subsequently retrained checkpoints. These results support task-specific readout and checkpoint routing as a simple way to recover epitope gains while retaining joint interface supervision.

1 Introduction

An antibody–antigen interface can be represented by a sparse binary matrix $Y \in \{0,1\}^{L_{\mathrm{Ab}} \times L_{\mathrm{Ag}}}$, where $Y_{ij} = 1$ marks a residue-pair contact. Paratope and epitope labels are deterministic row and column marginals of the same contact matrix. This relation makes joint learning attractive, but it also creates a practical model-selection problem: antigen-residue ranking and preservation of pair-level contacts or antibody-side residues can favor different checkpoints.

Recent methods have made epitope prediction antibody-aware using bipartite graph learning, multimodal interaction estimation, or interleaved cross-attention [1, 7, 14]. We study these design choices under one frozen SAbDab2 protocol and ask: *does the epitope gap arise from early optimization, late overfitting, or competition with contact and paratope supervision?* We then ask whether an epitope-specific readout can improve antigen-side prediction while retaining the stronger contact and paratope behavior of the joint model.

Our contributions are:

- We formulate contact, paratope, and epitope prediction around one explicit residue-pair contact map, combining pretrained sequence features with local N/CA/C backbone geometry and interleaved bidirectional cross-attention.
- A controlled 2×2 task ablation and epoch-level analysis produce curves consistent with late overfitting and do not support simple multitask competition, motivating us to retain the fully joint objective.
- We introduce a single-factor TopK2 epitope readout and deterministic task routing. The route improves epitope prediction on a previously unused, antigen-cluster-disjoint development holdout while leaving contact and paratope exactly unchanged.

- We bind data, candidate grids, partner panels, checkpoints, and seeds with immutable manifests across screening, a cluster-disjoint route confirmation, full-development retraining, and a disclosed non-blind historical measurement.

2 Related work

Antibody-aware interface prediction. AsEP provides a clustered benchmark and casts antibody-specific epitope prediction as bipartite link prediction; its WALLE baseline combines protein language models with structural graphs [7]. MIPE jointly predicts paratopes and epitopes using multimodal contrastive learning and an estimated interaction matrix [14]. EpiFormer interleaves bidirectional cross-attention inside geometric encoders and combines it with sparsity-aware objectives [1]. SPPIDER-seq demonstrates the value of partner-conditioned cross-attention in general protein-interface prediction [11]. Our EpiFormer-inspired component is a controlled readout adaptation inside our own 3D pair-contact architecture; the original EpiFormer and this adaptation have distinct model graphs.

Sequence and structure priors. Protein language models such as ESM-2 encode evolutionary regularities into residue features [6]. Geometry-aware graph networks provide useful inductive biases for molecular coordinates [13]. Paraplume shows that PLM representations alone can be strong and scalable for antigen-agnostic paratope prediction [2]. Our setting is instead partner-specific: antibody and antigen are both inputs, while cross-chain ground-truth distances serve exclusively as supervision. Because interface performance can differ between bound and unbound structural inputs [4], we explicitly disclose our bound-conformation setting below rather than compare it as if it were sequence-only.

3 Method

3.1 Geometry-aware pair-contact backbone

For each chain, external ESM-2 embeddings [6] are projected to $d = 128$. Separate antibody and antigen encoders combine a dense sequence path with a local spatial path constructed from N/CA/C residue frames, radial basis distance features, sequence offsets, and chain identity. Geometry enters through a zero-initialized residual gate, so the initial network remains close to the sequence pathway. Antibody and antigen use separate encoder parameters.

Backbone coordinates are extracted from deposited antibody–antigen complex structures. The model retains only each chain's intra-chain N/CA/C geometry: no cross-chain relative coordinates, native complex pose, or true cross-chain distance is supplied as input. Nevertheless, the intra-chain conformations are bound-complex-derived. The reported setting is therefore bound-conformation structural prediction, not sequence-only or unbound-monomer prediction.

Let h_i^{Ab} and h_j^{Ag} denote the resulting embeddings. The same directional low-rank decoder is applied in both chain orders:

$$\begin{aligned} \phi(x, y) &= \frac{q(x)^\top k(y)}{\sqrt{d_p}} + u(x) + v(y) + b, \\ z_{ij} &= \tfrac{1}{2}\left[\phi(h_i^{\mathrm{Ab}}, h_j^{\mathrm{Ag}}) + \phi(h_j^{\mathrm{Ag}}, h_i^{\mathrm{Ab}})\right], \quad p_{ij} = \sigma(z_{ij}), \end{aligned} \tag{1}$$

where $d_p = 128$. The head is evaluated over the complete antibody–antigen Cartesian candidate grid. At each interleaved fusion layer, antibody-to-antigen attention uses content logits only:

$$a_{ij}^{h,\mathrm{Ab}\leftarrow\mathrm{Ag}} = \underset{j:M_{ij}=1}{\mathrm{softmax}}\left(\frac{(W_q^h h_i^{\mathrm{Ab}})^\top (W_k^h h_j^{\mathrm{Ag}})}{\sqrt{d_h}}\right), \tag{2}$$

with the analogous update in the reverse direction. Partner context is injected through a zero-initialized residual gate. In the frozen experiments, no function of z_{ij} enters the attention logits.

Pair supervision is read from the frozen binary contact cohort. Residue labels are derived strictly from the pair matrix,

$$y_i^{\mathrm{para}} = \bigvee_j Y_{ij}, \qquad y_j^{\mathrm{epi}} = \bigvee_i Y_{ij}, \tag{3}$$

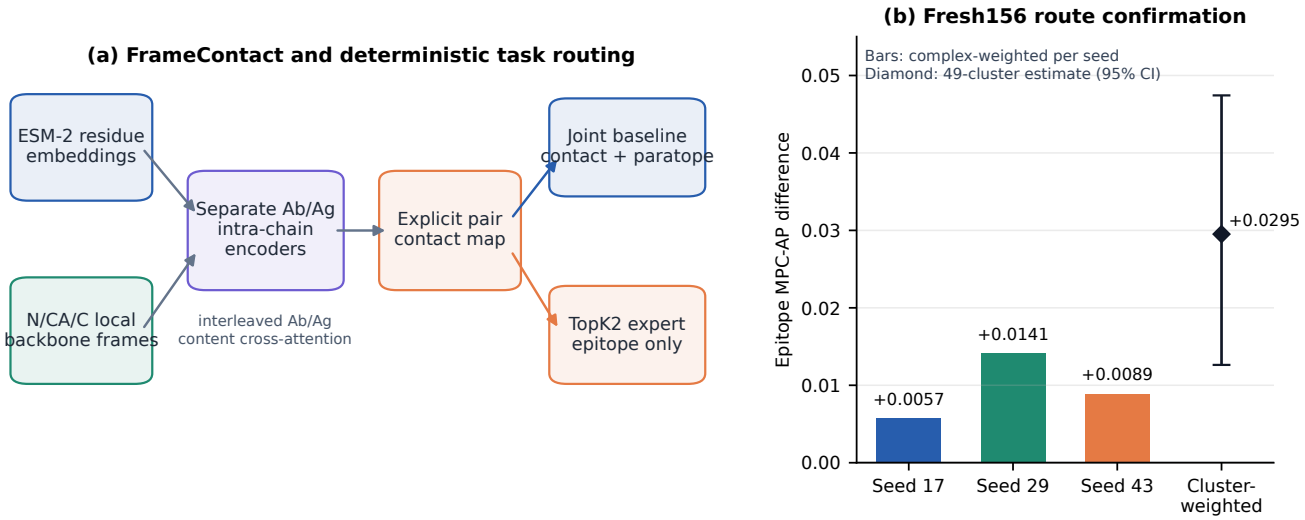


Figure 1: **Model and evidence flow.** FrameContact predicts an explicit residue-pair contact map from PLM embeddings and intra-chain backbone geometry. The joint baseline supplies contact and paratope; the TopK2 expert supplies epitope only. The route is a fixed task-to-checkpoint map. Right: two distinct fresh156 estimands—per-seed complex-weighted gains and the cluster-weighted, seed-averaged estimate with its conditional interval.

rather than imported surface annotations. The joint loss is

$$\mathcal{L} = \mathcal{L}_c + 0.1\mathcal{L}_d + 0.1\mathcal{L}_p + 0.5\mathcal{L}_e, \qquad \mathcal{L}_e = \mathcal{L}^e_{\text{CB-BCE}} + \mathcal{L}^e_{\text{Dice}} + 0.2\mathcal{L}^e_{\text{prev}}. \tag{4}$$

For each complex, binary losses are averaged within each observed class, then equally across observed classes; complexes also receive equal weight. $\mathcal{L}_c$ uses focal BCE with $\gamma = 1$ [5], whereas $\mathcal{L}_p$ and $\mathcal{L}^e_{\text{CB-BCE}}$ use BCE. With $\varepsilon = 10^{-6}$, the epitope-specific terms are

$$\begin{aligned} \mathcal{L}^e_{\text{Dice}} &= \frac{1}{B}\sum_b \left[1 - \frac{2\sum_r p_{br}y_{br} + \varepsilon}{\sum_r p_{br} + \sum_r y_{br} + \varepsilon}\right], \\ \mathcal{L}^e_{\text{prev}} &= \frac{1}{B}\sum_b \left|\frac{1}{n_b}\sum_r p_{br} - \frac{1}{n_b}\sum_r y_{br}\right|. \end{aligned} \tag{5}$$

$\mathcal{L}^e_{\text{Dice}}$ follows the differentiable Dice objective [9]. $\mathcal{L}_d$ is class-balanced cross-entropy on Cα–Cα distances bucketed at $4, 8, 16$, and 32 Å into five classes; it is averaged equally over observed classes within each complex and is auxiliary only.

3.2 Contact-derived epitope expert

Mean-like marginalization can dilute the few high-confidence edges of a sparse interface. For antigen residue j, the TopK2 expert instead computes

$$\begin{aligned} I_j &= \{i : M_{ij} = 1\}, & k_j &= \min(2, |I_j|), \\ \mu_j &= \frac{1}{k_j}\sum_{i \in \text{TopK}_{k_j}(\{p_{ij}: i \in I_j\})} p_{ij}, & s^{\text{epi}}_j &= \text{logit}(\text{clip}(\mu_j, \epsilon, 1-\epsilon)), \end{aligned} \tag{6}$$

where ϵ is float32 machine epsilon. Empty candidate sets map to zero logit, although the complete Cartesian grid prevents this case for real antigen residues. This is the only architectural factor changed from the same-protocol baseline; the joint contact/paratope/epitope supervision, loss weights, candidate grid, and partner panels remain fixed.

3.3 Deterministic task routing

Raw TopK2 checkpoints showed an epitope–paratope trade-off. We therefore freeze a per-task route:

$$\left(\hat{P}^c_r, \hat{p}^{\text{para}}_r, \hat{p}^{\text{epi}}_r\right) = \left(f^c_{\theta_{J,r}}(x), f^p_{\theta_{J,r}}(x), f^e_{\theta_{E,r}}(x)\right), \qquad r \in \{17, 29, 43\}, \tag{7}$$

where $\theta_{J,r}$ is the seed-matched joint baseline checkpoint and $\theta_{E,r}$ is the seed-matched TopK2 expert selected on development data. The route assigns one frozen checkpoint to each output using fixed task identity; the routed quantities are continuous scores, not thresholded labels.

4 Experimental protocol

Data and splits. We use a 3,663-complex SAbDab2 structural cohort [3] with a locally frozen, label-free antigen-fold split; it is not an official SAbDab2 split. Antigen components combine Foldseek reciprocal-min TM-score edges at 0.5 with exact PDB-assembly, antigen-sequence, and antibody-sequence identity; seed 20260730 assigns 2,565/549/549 complexes to train, validation, and the historical partition with zero exact-identity overlap across roles. Pair labels come from a frozen binary contact cache over the complete Cartesian candidate grid (190,509,766 pairs; 143,756 positives). The cache receipt authenticates the binary labels but does not independently record the upstream atom-level cutoff or missing-residue policy, so we do not infer either definition from the cache. All compared development arms use the same labels, candidate grid, external embeddings, and 31-way reciprocal partner panels.

Evidence stages. The 512/128 screen is used for task ablation and route design. Before measuring fresh156, we freeze the route and remove every instance and antigen cluster used by earlier selection or confirmation screens, leaving 156 complexes in 49 clusters. This measurement evaluates the frozen screen-stage route. We then retrain the expert on all 2,565 training complexes and select its checkpoint on the 549-complex validation set. Consequently, the full-development table is a validation summary, while fresh156 is independent evidence for the preceding routing decision rather than for the retrained checkpoint. The previously exposed 549-complex partition is reported only as a measurement-only historical appendix result.

Optimization and selection. We use three seeds (17, 29, 43), bf16, two self-geometry layers, eight heads, dropout 0.1, AdamW [8] with learning rate 3×10^{-4} and weight decay 10^{-4}, and batch size one. Checkpoints maximize validation epitope MPC-AP, break ties by contact MPC-AP and then earlier epoch, and use at most 20 epochs with patience 5. The full validation split supplies checkpoint selection, while fresh156 supplies the earlier antigen-cluster-disjoint route confirmation.

Metrics and statistics. We implement "AUPRC" as scikit-learn's non-interpolated average precision (AP; `average_precision_score`), not trapezoidal PR area [10]:

$$\text{AP} = \sum_k (R_k - R_{k-1}) P_k. \tag{8}$$

Because positives are sparse and complex sizes vary, the primary metric for task t averages AP only over positive-eligible complexes $\mathcal{E}_t = \{n : \sum_\ell y^{(t)}_{n\ell} > 0\}$:

$$\text{MPC-AP}_t = \frac{1}{|\mathcal{E}_t|} \sum_{n \in \mathcal{E}_t} \text{AP}(y_n, s_n), \tag{9}$$

with PR metrics preferred to ROC-only summaries under imbalance [12]. Pooled metrics retain all labeled examples, including those belonging to a zero-positive complex. Let δ_{rn} be the routed-minus-joint AP difference for seed r and complex n, and let C_g contain the complexes in antigen cluster g. We report both

$$\Delta_{\text{seed-med}} = \underset{r}{\text{median}} \frac{1}{N} \sum_n \delta_{rn}, \qquad \Delta_{\text{cluster}} = \frac{1}{|G|} \sum_{g \in G} \frac{1}{|C_g|} \sum_{n \in C_g} \frac{1}{3} \sum_r \delta_{rn}. \tag{10}$$

For fresh156, the confidence interval for Δ_{cluster} is the 2.5/97.5 percentile interval from 10,000 paired bootstrap resamples of the 49 antigen clusters after first averaging the three fixed seeds. This interval is conditional on those

trained seeds and does not include retraining uncertainty. The frozen promotion gate required median epitope gain ≥ 0.005, at least two positive seeds, minimum seed gain ≥ -0.01, and exact preservation of routed contact and paratope metrics.

5 Results

5.1 Joint learning remains beneficial

Both the same-protocol baseline and EpiFormer-inspired model peak early: the median best epoch across seeds is 2. Training loss continues to fall afterward while validation epitope AP declines. These curves are consistent with late overfitting, but do not by themselves identify a unique cause. The controlled task ablation in Table 1 supports retaining the fully joint objective, which is strongest; adding contact to epitope-only training is also mildly beneficial.

Table 1: Three-seed median epitope MPC-AP on the frozen 512/128 development screen. Architecture, split, and seed set are fixed.

Supervised outputs	Epitope MPC-AP
Epitope only	0.2502
Epitope + paratope	0.2493
Epitope + contact	0.2509
Epitope + paratope + contact	**0.2532**

5.2 Task routing transfers to an unseen development holdout

On fresh156, the routed model improves epitope MPC-AP for every seed (Table 2). The paired antigen-cluster bootstrap point estimate is +0.0295 with 95% CI $[0.0126, 0.0474]$ for the cluster-weighted, seed-averaged estimand in Eq. (10). Contact and paratope are exactly unchanged because those outputs are bound to the baseline checkpoint.

Table 2: Fresh156 epitope MPC-AP for the screen-stage checkpoints. The route was frozen before labels from this antigen-cluster-disjoint panel were measured; these are not evaluations of the later full-development retraining.

Seed	Joint baseline	Routed	Difference
17	0.1858	0.1915	+0.0057
29	0.1895	0.2036	+0.0141
43	0.1849	0.1938	+0.0089
Median difference			**+0.0089**

5.3 Full-development retraining and validation summary

The complete 2,565/549 development rerun passes every frozen promotion check. Two seeds improve and one regresses modestly; the three-seed median epitope MPC-AP increases by 0.0051. Table 3 reports the checkpoints selected on the same 549-complex validation set; it is not an independent confirmation. Contact and paratope are preserved exactly by routing.

Table 3: Full-development validation MPC-AP. Validation supplies checkpoint selection. Fresh156 independently supports the earlier route decision, not these retrained checkpoints.

Metric	Joint baseline	Routed	Difference
Contact	0.02858	0.02858	0.00000
Epitope	0.23241	**0.23746**	**+0.00505**
Paratope	0.70475	0.70475	0.00000

The per-seed epitope differences are +0.0130, +0.0051, and −0.0071. The median remains positive despite seed-level variation, and the result meets the predeclared median, positive-seed-count, and worst-seed criteria.

6 Discussion and conclusion

The curves are consistent with late overfitting, and the task ablation does not support simple contact/paratope competition as an explanation for the epitope gap. Contact-derived TopK2 aggregation plus deterministic routing produces a consistent epitope gain on fresh156 and a positive three-seed median gain on full development while preserving contact and paratope exactly. The one regressing full-development seed motivates stronger optimization stability and regularized checkpoint selection.

The same frozen protocol provides a direct path to two extensions: replacing ESM-2 with ESM-C to measure representation quality, and training external baselines on the identical split, candidate grid, and aggregation rules. A strictly later temporal cohort can then quantify performance under dataset shift. Together, these experiments position task-specific readout and routing as a simple, auditable addition to joint antibody–antigen interface learning. The principal limitations are the bound-complex-derived intra-chain geometry, the absence of an independent evaluation of the final retrained checkpoints, and the cache-level provenance boundary on the atom-level contact definition.

Reproducibility statement

Each checkpoint has 2,395,374 parameters. Routing evaluates two checkpoints, therefore requiring two forward passes. The expert reruns used one RTX 4090 per seed, bf16, batch size one, and 21.33 aggregate GPU-hours. Frozen ESM-2 input features use `facebook/esm2_t33_650M_UR50D`, revision `08e4846e5371`, 1,280 dimensions, and weight SHA-256 prefix `a08adabb949f`. The embedding-manifest SHA-256 prefix is `9ecbc0881516`.

The full-development protocol and analyzer SHA-256 prefixes are `a00a51a4617a` and `f4368275e38d`; full digests are stored in the accompanying machine-readable record. The split and binary-contact authorities are also hashed there. The accompanying machine-readable metrics file records the compact metrics, seeds, split roles, and audit status used in this draft. The archive also includes a machine-readable audit of every bibliographic record.

References

[1] Mansoor Ahmed, Huirong Chai, Haoxin Wang, Hemanth Venkateswara, and Murray Patterson. EpiFormer: Learning antigen–antibody interactions for epitope prediction via geometric deep learning. *arXiv preprint arXiv:2606.04154*, 2026. URL https://arxiv.org/abs/2606.04154.

[2] Gabriel Athènes, Adam Woolfe, Thierry Mora, and Aleksandra M. Walczak. Paraplume: A fast and accurate paratope prediction method provides insights into repertoire-scale binding dynamics. *arXiv preprint arXiv:2510.05626*, 2025. URL https://arxiv.org/abs/2510.05626.

[3] Henriette L. Capel, Odysseas Vavourakis, Benjamin H. Williams, Christopher R. Taylor, and Charlotte M. Deane. SAbDab2: The structural antibody database in the age of machine learning. *bioRxiv*, 2026. doi: 10.64898/2026.06.16.732554. URL https://www.biorxiv.org/content/10.64898/2026.06.16.732554v1.

[4] Ana Davila, Zichang Xu, Songling Li, John Rozewicki, Jan Wilamowski, Sergei Kotelnikov, Dima Kozakov, Shunsuke Teraguchi, and Daron M. Standley. AbAdapt: An adaptive approach to predicting antibody–antigen complex structures from sequence. *Bioinformatics Advances*, 2(1):vbac015, 2022. doi: 10.1093/bioadv/vbac015.

[5] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2980–2988, 2017. URL https://openaccess.thecvf.com/content_iccv_2017/html/Lin_Focal_Loss_for_ICCV_2017_paper.html.

[6] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido,

and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

[7] Chu'nan Liu, Lilian Denzler, Yihong Chen, Andrew Martin, and Brooks Paige. AsEP: Benchmarking deep learning methods for antibody-specific epitope prediction. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0373.

[8] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

[9] Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-Net: Fully convolutional neural networks for volumetric medical image segmentation. In *2016 Fourth International Conference on 3D Vision*, pages 565–571, 2016. doi: 10.1109/3DV.2016.79.

[10] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011. URL https://www.jmlr.org/papers/v12/pedregosa11a.html.

[11] Aleksey Porollo, Om Jadhav, Aaron Alvarez, and Jichao Chen. SPPIDER-seq: Sequence-based partner-aware predictor of protein–protein interaction sites. *Bioinformatics*, 42(7):btag388, 2026. doi: 10.1093/bioinformatics/btag388.

[12] Takaya Saito and Marc Rehmsmeier. The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015. doi: 10.1371/journal.pone.0118432.

[13] Víctor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 9323–9332, 2021. URL https://proceedings.mlr.press/v139/satorras21a.html.

[14] Zhiwei Wang, Yongkang Wang, and Wen Zhang. Improving paratope and epitope prediction by multi-modal contrastive learning and interaction informativeness estimation. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, pages 6053–6061, 2024. doi: 10.24963/ijcai.2024/669.

A Historical SAbDab2 benchmark

For transparency, Table 4 reports a previously exposed, single-seed, measurement-only historical benchmark. This partition had already been accessed, and prior test feedback influenced later development. The present measurement did not select the route, model, checkpoint, threshold, or calibration, but it is not a fresh blind test and cannot confirm generalization. Thresholds were selected on development validation and were not refit here. Pooled metrics use all 549 complexes; per-complex AP excludes one zero-positive complex and therefore averages over 548 eligible complexes.

Table 4: Previously exposed, non-blind, single-seed historical measurement; not used for model selection. AP columns are non-interpolated average precision.

Task	AUROC	AP	F1	MCC	MPC-AP
Contact	0.9371	0.0121	0.0370	0.0519	0.0264
Epitope	0.6915	0.1411	0.2119	0.1564	0.2486
Paratope	0.9567	0.6567	0.6440	0.6097	0.7095